In [13]:
```
n = int(input())
if n%2 == 0:
    print("even")
else:
    print("odd")
```

```
even
```

In [3]:
```
n = int(input("Enter your age: "))
if n >= 18:
        print("elgible for vote")
else:
        print("not elgilble for vote")
```

```
elgible for vote
```

In [7]:
```
year = int(input())
if ( year%4 ==0 and year % 100 != 0) or year % 400 == 0:
    print("leap year")
else:
    print("not a leap year")
```

```
not a leap year
```

In [9]:
```
n = int(input())
if n == 0:
    print("zero")
elif n <0 :
    print("negative num")
else:
    print("positive num")
```

```
positive num
```

In [11]:
```
a,b = 10,20
if a>b :
    print("a is greater than b")
elif b>a:
    print("a is less than b")
else:
    print("a is equal to b")
```

```
a is less than b
```

In [12]:
```
n = int(input())
if n % 5 == 0:
    print("it is 5 multiple")
else:
    print("it is not 5 multiple")
```

```
it is 5 multiple
```

In [16]:
```
char = input()
if char in "aeiouAEIOU":
    print("vowel")
else:
    print("consonant")
```

```
vowel
```

In [19]:
```
dig = int(input())
st = len(str(dig))
if st == 1:
    print("Singel digit num")
```

```
else:
    print("multi digit")
```

```
multi digit
```

In [21]:
```
time = int(input())
if time < 12:
  print("Good morning")
else:
    print("Good afteroon")
```

```
Good afteroon
```

In [22]:
```
st = input()
if not st:
    print("emty")
else:
    print("not empty")
```

```
not empty
```

In [24]:
```
import math
num = int(input("Enter a number: "))
if math.isqrt(num) ** 2 == num:
    print("Perfect Square")
else:
    print("Not a perfect square")
```

```
Perfect Square
```

In [25]:
```
num = int(input("Enter a number: "))
if 1 <= num <= 100:
    print("Within range")
else:
    print("Out of range")
```

```
Within range
```

In [26]:
```
day = input().lower()
if day in ['saturday','sunday']:
    print("weekend")
else:
    print("Weekday")
```

```
weekend
```

In [27]:
```
num = int(input("Enter a number: "))
if num % 3 == 0 and num % 7 == 0:
    print("Divisible by 3 and 7")
else:
    print("Not divisible by both")
```

```
Not divisible by both
```

In [28]:
```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
if (a + b) > 100:
    print("Sum is greater than 100")
else:
    print("Sum is 100 or less")
```

```
Sum is 100 or less
```

In [29]:
```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
if a < b:
    print("Minimum:", a)
else:
    print("Minimum:", b)
```

```
Minimum: 67
```

In [32]:

```python
num = int(input("Enter a number: "))
if num % 2 == 0 and num % 3 != 0:
    print("Divisible by 2 but not by 3")
else:
    print("Does not meet criteria")
```

```
Divisible by 2 but not by 3
```

In [34]:

```python
char = input()
if char.isupper():
    print("upper case")
else:
    print("lower case")
```

```
lower case
```

In [35]:

```python
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
if a >= b and a >= c:
    print("Largest:", a)
elif b >= a and b >= c:
    print("Largest:", b)
else:
    print("Largest:", c)
```

```
Largest: 89
```

In [44]:

```python
num = int(input())
if num > 1:
    for i in range(2,num):
        if num % i  == 0:
            print("not prime num")
            break
    else:
            print("prime num")
else:
    print("npt prime")
```

```
not prime num
```

In [45]:

```python
age = int(input("Enter age: "))
passed_test = input("Did you pass the driving test? (yes/no): ").lower()
if age >= 18 and passed_test == "yes":
    print("Eligible for a driving license")
else:
    print("Not eligible")
```

```
Eligible for a driving license
```

In [46]:

```python
a = int(input("Enter first side: "))
b = int(input("Enter second side: "))
c = int(input("Enter third side: "))
if a == b == c:
    print("Equilateral Triangle")
elif a == b or b == c or a == c:
    print("Isosceles Triangle")
else:
    print("Scalene Triangle")
```

```
Isosceles Triangle
```

In [47]:

```python
marks = int(input("Enter marks: "))
if marks >= 40:
```

```
    print("Pass")
else:
    print("Fail")
```

```
Pass
```

In [48]:
```
num = input("Enter a number: ")
if num == num[::-1]:
    print("Palindrome")
else:
    print("Not a Palindrome")
```

```
Palindrome
```

In [49]:
```
units = int(input("Enter electricity units consumed: "))
if units <= 100:
    bill = units * 5
elif units <= 300:
    bill = (100 * 5) + (units - 100) * 10
else:
    bill = (100 * 5) + (200 * 10) + (units - 300) * 15
print("Total Bill: ₹", bill)
```

```
Total Bill: ₹ 435
```

In [50]:
```
marks = int(input("Enter marks: "))
if marks >= 90:
    print("Grade: A")
elif marks >= 80:
    print("Grade: B")
elif marks >= 70:
    print("Grade: C")
elif marks >= 60:
    print("Grade: D")
elif marks >= 40:
    print("Grade: E")
else:
    print("Grade: F (Fail)")
```

```
Grade: E
```

In [52]:
```
import calendar
day = int(input("Enter day: "))
month = int(input("Enter month: "))
year = int(input("Enter year: "))
if 1 <= month <= 12 and 1 <= day <= calendar.monthrange(year, month)[1]:
    print("Valid date")
else:
    print("Invalid date")
```

```
Invalid date
```

In [53]:
```
hour = int(input("Enter hour (24-hour format): "))
if hour < 12:
    print("AM")
else:
    print("PM")
```

```
AM
```

In [55]:
```
num = input("Enter a number: ")
power = len(num)
if sum(int(digit) ** power for digit in num) == int(num):
    print("Armstrong Number")
else:
    print("Not an Armstrong Number")
```

Armstrong Number

In [57]:
```
a = int(input("Enter first side: "))
b = int(input("Enter second side: "))
c = int(input("Enter third side: "))
d = int(input("Enter fourth side: "))
if a == b == c == d:
    print("Square")
elif a == c and b == d:
    print("Rectangle")
else:
    print("Other Quadrilateral")
```

Other Quadrilateral

In [58]:
```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
op = input("Enter operation (+, -, *, /): ")
if op == "+":
    print("Result:", a + b)
elif op == "-":
    print("Result:", a - b)
elif op == "*":
    print("Result:", a * b)
elif op == "/":
    print("Result:", a / b)
else:
    print("Invalid operation")
```

Invalid operation

In [60]:
```
balance = float(input("Enter account balance: "))
withdraw = float(input("Enter withdrawal amount: "))
if balance >= withdraw:
    print("Withdrawal successful")
else:
    print("Insufficient funds")
```

Withdrawal successful

In [63]:
```
import re
password = input("Enter password: ")
if len(password) >= 8 and re.search(r"[A-Za-z]", password) and re.search(r"\d", password):
    print("Valid Password")
else:
    print("Invalid Password")
```

Invalid Password

In [65]:
```
month = int(input("Enter month (1-12): "))
year = int(input("Enter year: "))
days = [31, 28 + (1 if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0) else 0), 31,
30, 31, 30, 31, 31, 30, 31, 30, 31]
print("Days:", days[month - 1])
```

Days: 29

In [ ]: